

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8EJ402			
Course Title	PARTIAL DIFFERENTIAL EQUATIONS			
Type of Course	Elective			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial	Practical	Total Hours
		per week	per week	
	4	4	-	60
Pre-requisites	1. Real Analysis 2. Basic Concepts of Vector functions 2. Ordinary Differential Equations			
Course Summary	This introductory Partial Differential Equations (PDEs) course equips students with the mathematical tools and problem-solving skills necessary to analyse and solve real-world phenomena governed by PDEs. The syllabus focuses on analytical methods for solving first and second-order PDEs, laying the foundation for further exploration of advanced PDEs and their applications.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understanding of basic concepts, definitions, and mathematical problems related to first-order quasilinear equations.	U	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Analyse and evaluate the classification of second-order linear equations, including the Cauchy problem and wave equations.	An	E	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Evaluate solutions for boundary value problems and apply them in solving PDEs.	E	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text: Linear Partial Differential Equations for Scientists and Engineers, Fourth Edition, Tyn Myint-U, Lokenath Debnath, Birkhauser(2007), ISBN : 978-81-8489-079-2.				
Module	Unit	Content	Hrs (48 +12)	Ext. Marks (70
I	First Order Quasilinear Equations and Method of Characteristics		9	
	1	Basic Concepts, definitions and mathematical problems		
	2	Classification of first order equations		
	3	Construction of a first order equation		
	4	Geometrical Interpretation of a First- Order Equation		
	5	Method of characteristics and General solutions		
		Sections from Text: 1.2, 1.3, 2.1, 2.2,2.3, 2.4, 2.5.		
II	Classification of Second Order Linear Equations, The Cauchy Problem and Wave Equations		21	
	6	Second order equations in two independent variables		
	7	Canonical Forms		
	8	Equations with constant coefficients		
	9	General Solutions		
	10	The Cauchy Problem		
	11	Homogeneous Wave Equations		
	12	Initial Boundary-Value Problems		
	13	Equations with Nonhomogeneous Boundary Conditions		
	14	Vibration of Finite String with Fixed Ends		
	15	Nonhomogeneous Wave Equations		
	16	The Riemann Method		

	Sections from Text: 4.1 - 4.4, 5.1, 5.3-5.8		
III	Method of Separation of Variables		13
	17	Introduction	
	18	Separation of Variables	
	19	The Vibrating String Problem	
	20	Existence and Uniqueness of Solution of the Vibrating String Problem	
	21	The Heat Conduction Problem	
	22	Existence and Uniqueness of Solution of the Heat Conduction Problem	
	23	The Laplace and Beam Equations	
	24	Nonhomogeneous Problems	
	Sections from Text: 7.1-7.8		
IV	Boundary Value Problems and Applications		7
	25	Boundary Value Problems	
	26	Maximum and Minimum Principles	
	27	Uniqueness and Continuity Theorems	
	28	Dirichlet Problem for a circle	
	29	Neumann Problem for a circle	
	30	Dirichlet Problem for a rectangle	
	31	The Neumann Problem for a Rectangle	
	Sections from Text: 9.1-9.4, 9.6, 9.7, 9.8,9.9		
V (Open Ended)	Green's Functions, Boundary Value Problems and Nonlinear Equations		12
		Green's Functions for Ordinary Differential Equations, Construction of Green's Functions, The Dirac Delta Function, Properties of Green's Functions, Method of Green's Functions (only for Laplace operator) Nonlinear PDEs -brief overview from any text	

References:

1. Partial Differential Equations -An Introduction, Second Edition, Walter A. Strauss, John Wiley and Sons Limited.
2. Partial Differential Equations-Classical Theory with a Modern Touch, A.K. Nandakumaran, P.S. Datti, Cambridge-IISc Series.
3. Elements of Partial Differential Equations, I.N. Sneddon, McGraw-Hill, New York (1972).

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	3	1	2	1	3	0	1
CO 2	3	3	2	1	3	1	3	1	3	0	1
CO 3	2	3	2	1	3	1	3	1	3	0	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓